



SOP #16

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Antibody Incubation

Protocol summary:

Incubate (blocked) blots with primary antibody (1 hour)
Rinse protocol (30 min)
Incubate blots with secondary antibody (1 hour)
Rinse protocol (30 min)
Visualize using CCD camera & chemiluminescent detection agent
Analysis with ImageLab software

Important Note: All blots should be labelled on the box (not the lid) with initials, date, and target.

1. Pour new 10 mL of blocking solution (2% ECL block agent or 5% milk powder) over the membrane in each immunodetection box.
2. Add the appropriate volume of primary antibody to the blocking solution in each of the immunodetection boxes (note the product number and lot number). Rinse out the pipette tip with the blocking solution. Rock the immunodetection boxes gently to mix. Incubate the membrane with the primary antibody for 1 h at room temperature with gentle constant agitation.
3. Each blot box holds 10,000 uL (10 mL), calculate how to make an intermediate dilution in 1XTBS-T. Dilutions are good for one week in the fridge. Ex: prepare a 1/10 dilution of antibody by adding 2 uL antibody sera to 18 uL of 1X TBS-T, then add 10 uL to the 10,000 uL of blocking agent. This is the equivalent of 1/10,000 antibody dilution.

$$\frac{2}{20} \times \frac{10}{10,000} = \frac{20}{200,000} = \frac{1}{10,000}$$

4. After 1 hr incubation, wash each membrane with 15 to 20 mL of 1XTBS-T:
 - a. Twice briefly.
 - b. Once for 15 min.
 - c. Thrice for 5 min each.
5. Pour 10 mL of blocking solution over the membrane in each immunodetection box.
6. Add the appropriate volume of secondary antibody to the blocking solution in each immunodetection box, following similar protocol as for primary. Incubate the membrane with the secondary antibody for 1 h at room temperature with gentle constant agitation.
7. Wash each membrane with protocol from step 3.
8. Take out the detection reagents from the refrigerator. Allow to equilibrate to room temperature before using ($\approx$ 30 min).